TEXORA · LECTURE SHEET

Flat Knitting

Two flat beds inclined as an inverted V, a carriage that carries cams and yarn across them, and needles that can knit, tuck, miss, transfer and rack: the V-bed flat machine trades the circular machine's speed for something circular machines cannot do — shape the fabric, panel by panel, even to a complete garment.

TE-3101 · KNITTING TECHNOLOGY · TOPIC 10 · LEVEL 3 TERM 1

1. The V-bed arrangement

Front and back beds are flat plates of needle tricks, inclined at about 90° to each other (each ≈ 45° from horizontal), meeting at the top in a narrow gap — the inverted V. Latch needles slide in the tricks; a **carriage** traversing the beds carries the cam systems (usually two or three systems) and yarn feeders, knitting in both directions of travel. Above the gap, **brushes** open latches, a **sinker/loop-presser arrangement** and modern **stitch-presser / holding-down devices** control the fabric at the gap, and takedown rollers below draw the panel down between the beds. Bed width (e.g. 36–84 in) and gauge (E3–E18 for classic knitwear; finer for whole-garment machines) define the product class.

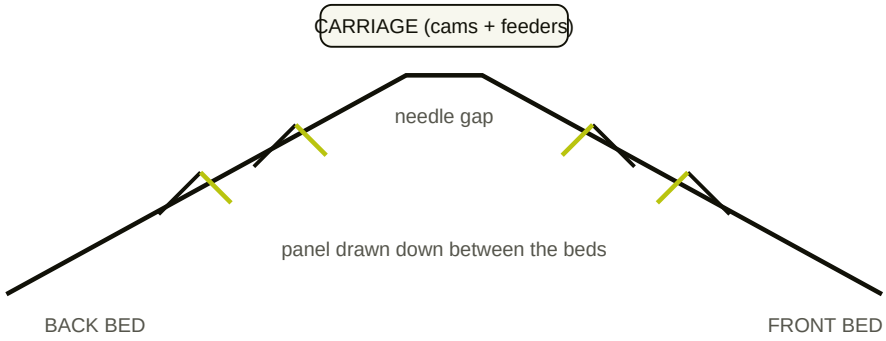


Figure 1 — V-bed geometry. Needles from opposite beds can pass each other at the gap — the condition for rib knitting and for loop transfer between beds.

2. Cam systems and needle selection on the carriage

Each carriage system repeats the Topic 3 logic in flat form — clearing cams, stitch cams (independently settable per bed and per carriage direction, some machines varying stitch length within a course), tuck/miss selection — but the programme is per needle: modern machines select every needle electronically each course (knit/tuck/miss/transfer), giving jacquard, intarsia (separate feeders per

colour block, carried by yarn carriers that park at their block edges), cables (loop transfer across neighbouring needles, aided by **racking** — shifting one bed sideways by one or more pitches), and lace transfer patterns. Carriage speed is quoted in m/s (typically up to ≈ 1.2–1.6 m/s); courses per minute depend on working width — the narrower the panel, the more traverses per minute, which is why flat knitting economics are panel-based.

3. Shaping: fully-fashioned and whole-garment knitting

Fashioning changes the number of working wales: **widening** brings an idle needle into action at the edge; **narrowing** transfers edge loops inward onto adjacent needles (the transferred loop marks the fashioning point). Armholes, necklines, sleeve tapers and sock heels are all programmed sequences of transfers. **Whole-garment** machines (four-bed or compound-needle V-beds) knit body and sleeves as tubes and join them on the machine — no cutting, minimal sewing. **Worked check:** a front panel starts at 240 wales and narrows by 1 wale each side every 4 courses for 40 courses: reductions = 40 ÷ 4 = 10 per side; final wales = 240 − 2 × 10 = **220 wales** — the arithmetic every shaping programme is built from.

4. Structures on the flat machine

STRUCTURE	HOW	PRODUCT
Plain panels	Front bed only	Sweater fronts/backs (cut or fashioned)
Ribs 1×1, 2×2, cardigans	Both beds, gating by bed offset	Cuffs, welts, collars, full garments
Collars and trims	Short-width rib/interlock-type work, often on dedicated collar machines	Polo collars/cuffs for the circular-knit trade
Jacquard / intarsia	Per-needle selection; intarsia carriers per block	Patterned knitwear
Cables, lace, pointelle	Loop transfer ± racking	Fashion knitwear

5. Flat versus circular — the production choice

Circular wins on kg/h of commodity fabric (Topic 4: ~31 kg/h) and loses everything to cutting waste and sewing. Flat wins wherever shape, pattern or garment integration carries the value: a fashioned sweater panel leaves the machine garment-shaped, and waste is the yarn in selvedges, not cut-away

fabric. Factories therefore run both: circular for bodies of basic tees/polos, flat for collars, cuffs and all true knitwear.

Glossary

Carriage — the traversing unit carrying cams and feeders. **Racking** — lateral shifting of one bed relative to the other. **Fashioning** — shaping by adding/removing wales via transfer. **Intarsia** — colour blocks knitted with separate yarn carriers, no floats across the back. **Welt** — the starting band of a panel/garment.

Key points

• V-bed = two inclined flat beds + traversing carriage; per-needle electronic selection gives knit/tuck/miss/transfer at will.
• Transfer + racking = shaping, cables and lace — flat knitting's reason to exist.
• Economics are per panel/garment, not per kg; shaping arithmetic is wales gained/lost per fashioning step.

Practice

1. A sleeve starts at 120 wales, widens 1 wale per side every 6 courses for 66 courses. Find the final wale count.
2. Explain why intarsia has no long floats on the back while flat-bed jacquard does.
3. A carriage runs at 1.2 m/s over a 1.0 m working width. Estimate traverses (courses) per minute, ignoring reversal time, and say why real output is lower.
4. Choose machine type (circular S/J, circular rib, flat V-bed) for: (a) 500 kg of plain jersey, (b) 2000 polo collars, (c) 300 fashioned sweaters. Justify each in one line.

Answer key

1. Steps = 66 ÷ 6 = 11 per side; gain = 2 × 11 = 22; final = 120 + 22 = **142 wales**.
2. Intarsia gives each colour block its own carrier that travels only within its block, so colour areas join at their edges without yarn crossing the back. Jacquard selection knits one colour per course section from feeders spanning the full width, so unselected colours float behind the needles that did not take them.
3. Time per traverse = 1.0 m ÷ 1.2 m/s = 0.83 s → ≈ **72 courses/min** ideal. Real output is lower: carriage reversal and acceleration, transfer courses (slower), feeder changes and takedown/programme pauses all consume time.

4. (a) Circular S/J — volume tube fabric. (b) Flat V-bed (collar work) — short-width rib panels knitted to shape. (c) Flat V-bed — fashioning by transfer is only possible there.
